

Solved problems

**Gradient Descent, Chain Rule and Regularization
Techniques**

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Problem 1:

Consider the quadratic function of one variable:

$$f(x) = 3x^2 + 6x + 5.$$

(a) Compute the derivative $f'(x)$.

(b) Is the function convex? Justify your answer.

(c) Perform **two iterations** of gradient descent starting from the initial point $x_0 = 2$ with a learning rate $\alpha = 0.1$.

(d) Find the exact minimizer x^* by solving $f'(x) = 0$. Also compute the minimum value $f(x^*)$.

(e) Compare the point obtained after two iterations with the exact minimizer. Compute the absolute error $|x_2 - x^*|$.

For a quadratic function, the error in gradient descent reduces by a factor of $|1 - \alpha f''|$ each iteration, where f'' is the constant second derivative. Use this to estimate how many additional iterations would be needed to get within an error of 0.001 of the exact minimizer.

(f) Suppose you increase the learning rate to $\alpha = 0.4$. Perform only the **first iteration** from $x_0 = 2$. What is the new point? Compute the function value at that point and compare it with the starting value. Is this learning rate likely to lead to convergence? Explain briefly.

Solution**(a) Derivative**

$$f'(x) = 6x + 6.$$

(b) Convexity

The second derivative is constant: $f''(x) = 6$. Since $f'' > 0$ for all x , the function is strictly convex (it has a unique global minimum).

(c) Two Iterations of Gradient Descent with $\alpha = 0.1$ **Iteration 1 (k = 0):**

Current point: $x_0 = 2$

Gradient: $f'(2) = 6(2) + 6 = 12 + 6 = 18$.

Update:

$$x_1 = x_0 - \alpha f'(x_0) = 2 - 0.1 \times 18 = 2 - 1.8 = 0.2.$$

Function value:

$$f(0.2) = 3(0.2)^2 + 6(0.2) + 5 = 3(0.04) + 1.2 + 5 = 0.12 + 1.2 + 5 = 6.32.$$

Iteration 2 (k = 1):

Current point: $x_1 = 0.2$

Gradient: $f'(0.2) = 6(0.2) + 6 = 1.2 + 6 = 7.2$.

Update:

$$x_2 = 0.2 - 0.1 \times 7.2 = 0.2 - 0.72 = -0.52.$$

Function value:

$$f(-0.52) = 3(0.2704) + 6(-0.52) + 5 = 0.8112 - 3.12 + 5 = 2.6912.$$

(d) Exact Minimizer

Set $f'(x) = 0$:

$$6x + 6 = 0 \Rightarrow 6x = -6 \Rightarrow x^* = -1.$$

Minimum value:

$$f(-1) = 3(1) + 6(-1) + 5 = 3 - 6 + 5 = 2.$$

(e) Error and Convergence Estimate

After two iterations, $x_2 = -0.52$. The absolute error is

$$|x_2 - x^*| = |-0.52 - (-1)| = |0.48| = 0.48.$$

For a quadratic, the second derivative $f'' = 6$ is constant. With learning rate $\alpha = 0.1$, the error reduction factor per iteration is

$$\gamma = |1 - \alpha f''| = |1 - 0.1 \times 6| = |1 - 0.6| = 0.4.$$

Thus after each iteration, the error multiplies by 0.4. Let e_k be the error after k additional iterations, starting from current error $e_0 = 0.48$. Then

$$e_k = 0.48 \times (0.4)^k.$$

We want $e_k < 0.001$:

$$0.48 \times (0.4)^k < 0.001 \Rightarrow (0.4)^k < \frac{0.001}{0.48} \approx 0.0020833.$$

Take natural logs:

$$k \ln(0.4) < \ln(0.0020833) \Rightarrow k > \frac{\ln(0.0020833)}{\ln(0.4)}.$$

$\ln(0.0020833) \approx -6.1738$, $\ln(0.4) \approx -0.9163$. So

$$k > \frac{-6.1738}{-0.9163} \approx 6.74.$$

Thus approximately **7 additional iterations** are needed (total 9 from start) to get within 0.001 of the exact minimum.

(f) Larger Learning Rate $\alpha = 0.4$

Perform first iteration from $x_0 = 2$ with $\alpha = 0.4$:

Gradient at x_0 : $f'(2) = 18$.

Update:

$$x_1 = 2 - 0.4 \times 18 = 2 - 7.2 = -5.2.$$

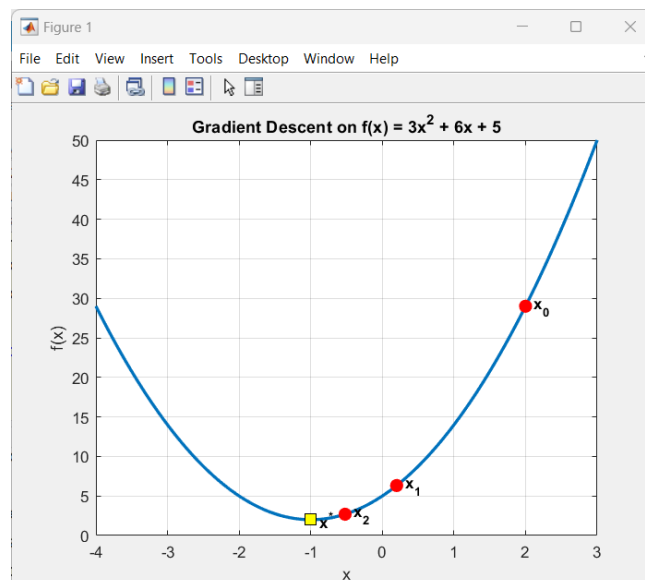
Function value:

$$f(-5.2) = 3(27.04) + 6(-5.2) + 5 = 81.12 - 31.2 + 5 = 54.92.$$

Starting value $f(2) = 3(4) + 12 + 5 = 12 + 12 + 5 = 29$. The function value increased dramatically from 29 to 54.92, indicating divergence. The learning rate is too large; the step overshoots the minimum and moves to a point where the function is higher. This rate is unlikely to lead to convergence (it will likely oscillate and diverge).

Summary table of gradient descent iterations:

Iteration	x	f_x
0	2	29
1	0.2	6.32
2	-0.52	2.6912



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Problem 2:

Consider the multivariate function

$$f(x, y) = 2x^2 + 3y^2 - 2xy + 4x - 5y.$$

(a) Compute the gradient vector $\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$.

(b) Find the Hessian matrix H of f . Is f convex? Justify your answer by examining the Hessian.

(c) Perform **two iterations** of gradient descent starting from the initial point $(x_0, y_0) = (1, 1)$ with a learning rate $\alpha = 0.1$.

(d) Find the exact minimizer (x^*, y^*) of f by solving the system $\nabla f(x, y) = 0$. Also compute the minimum value $f(x^*, y^*)$.

(e) Compare the point obtained after two iterations with the exact minimizer. Compute the Euclidean distance between them. Based on the rate of convergence, estimate how many additional iterations might be needed to get within a distance of 0.01 from the exact minimizer. (Hint: For a quadratic function, the error reduces roughly by a constant factor each step; you can approximate using the largest and smallest eigenvalues of the Hessian.)

(f) Suppose you increase the learning rate to $\alpha = 0.5$. Perform only the **first iteration** from the same starting point $(1, 1)$. What is the new point? Compute the function value at that point and compare it with the starting value. Is this learning rate likely to lead to convergence? Explain briefly.

Solution:

(a) Gradient Computation

$$\frac{\partial f}{\partial x} = 4x - 2y + 4, \frac{\partial f}{\partial y} = 6y - 2x - 5.$$

Thus,

$$\nabla f(x, y) = (4x - 2y + 4, 6y - 2x - 5).$$

(b) Hessian and Convexity

The Hessian matrix is the matrix of second partial derivatives:

$$H = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ -2 & 6 \end{pmatrix}.$$

To check convexity, we examine the eigenvalues of H . Solve $\det(H - \lambda I) = 0$:

$$\det \begin{pmatrix} 4 - \lambda & -2 \\ -2 & 6 - \lambda \end{pmatrix} = (4 - \lambda)(6 - \lambda) - 4 = \lambda^2 - 10\lambda + 20 = 0.$$

The eigenvalues are $\lambda = \frac{10 \pm \sqrt{100-80}}{2} = 5 \pm \sqrt{5}$. Numerically, $\sqrt{5} \approx 2.236$, so $\lambda_1 \approx 7.236$, $\lambda_2 \approx 2.764$.

Both are positive. Since H is symmetric and positive definite (all eigenvalues > 0), f is strictly convex. (A convex function has a unique global minimum.)

(c) Two Iterations of Gradient Descent with $\alpha = 0.1$

Iteration 1 ($k = 0$):

Current point: $(x_0, y_0) = (1, 1)$

Update:

$$(x_1, y_1) = (x_0, y_0) - \alpha \nabla f(x_0, y_0) = (1, 1) - 0.1 \times (6, -1) = (1 - 0.6, 1 + 0.1) = (0.4, 1.1).$$

Function value at new point:

$$f(0.4, 1.1) = 2(0.4)^2 + 3(1.1)^2 - 2(0.4)(1.1) + 4(0.4) - 5(1.1) = -0.83$$

Iteration 2 ($k = 1$):

Current point: $(x_1, y_1) = (0.4, 1.1)$

Update:

$$(x_2, y_2) = (0.4, 1.1) - 0.1 \times (3.4, 0.8) = (0.06, 1.02).$$

Function value:

$$f(0.06, 1.02) = 2(0.06)^2 + 3(1.02)^2 - 2(0.06)(1.02) + 4(0.06) - 5(1.02) = -1.854$$

(d) Exact Minimizer

Set gradient to zero:

$$\begin{cases} 4x - 2y + 4 = 0 \Rightarrow 4x - 2y = -4 \Rightarrow 2x - y = -2 & (1) \\ -2x + 6y - 5 = 0 \Rightarrow -2x + 6y = 5 & (2) \end{cases}$$

Solve (1) for y : $y = 2x + 2$. Substitute into (2):

$$-2x + 6(2x + 2) = 5 \rightarrow -2x + 12x + 12 = 5 \rightarrow 10x = -7 \rightarrow x^* = -0.7.$$

$$\text{Then } y^* = 2(-0.7) + 2 = -1.4 + 2 = 0.6.$$

So exact minimizer: $(x^*, y^*) = (-0.7, 0.6)$.

Minimum value:

$$f(-0.7, 0.6) = 2(0.49) + 3(0.36) - 2(-0.42) + 4(-0.7) - 5(0.6)$$

$$2(0.49) = 0.98$$

$$3(0.36) = 1.08$$

$$-2(-0.42) = +0.84$$

$$4(-0.7) = -2.8$$

$$-5(0.6) = -3.0$$

Sum: $0.98 + 1.08 = 2.06$; $2.06 + 0.84 = 2.90$; $2.90 - 2.8 = 0.10$; $0.10 - 3.0 = -2.90$.

So $f(x^*, y^*) = -2.90$.

(e) Comparison and Convergence Estimate

After two iterations, we have $(x_2, y_2) = (0.06, 1.02)$. The Euclidean distance to the exact minimizer $(-0.7, 0.6)$ is:

$$d = \sqrt{(0.06 - (-0.7))^2 + (1.02 - 0.6)^2} \approx 0.868.$$

For a quadratic function, gradient descent with a fixed learning rate converges linearly.

The convergence rate depends on the eigenvalues of the Hessian.

The eigenvalues are $\lambda_{\min} \approx 2.764$ and $\lambda_{\max} \approx 7.236$.

For a convex quadratic, the optimal learning rate is $\alpha_{\text{opt}} = \frac{2}{\lambda_{\min} + \lambda_{\max}} \approx \frac{2}{10} = 0.2$. Our

learning rate $\alpha = 0.1$ is half the optimal, so the error reduction factor per iteration is approximately $\max(|1 - \alpha\lambda_{\min}|, |1 - \alpha\lambda_{\max}|)$. Compute:

$$1 - \alpha\lambda_{\min} = 1 - 0.1 \times 2.764 = 1 - 0.2764 = \mathbf{0.7236},$$

$$1 - \alpha\lambda_{\max} = 1 - 0.1 \times 7.236 = 1 - 0.7236 = 0.2764.$$

The error reduction is dominated by the larger factor, which is **0.7236**.

Thus, after each iteration, the distance to the minimum roughly multiplies by about **0.724**.

We want the distance to be less than 0.01. Starting from $d_0 \approx \mathbf{0.868}$ after two iterations (we can treat this as current error). Let d_k be the error after k additional iterations. Then

$$d_k \approx \mathbf{0.868} \times (\mathbf{0.724})^k.$$

We need $0.868 \times (0.724)^k < 0.01 \rightarrow (0.724)^k < 0.01152$. Take natural logs:

$$k \ln(0.724) < \ln(0.01152) \Rightarrow k > \frac{\ln(0.01152)}{\ln(0.724)}.$$

$\ln(0.01152) \approx -4.463$, $\ln(0.724) \approx -0.322$. So

$$k > \frac{-4.463}{-0.322} \approx 13.86.$$

Thus approximately **14 additional iterations** are needed (total 16 iterations from start) to get within 0.01 of the exact minimizer.

MATLAB results

Function $f(x,y) =$
 $2x^2 - 2xy + 4x + 3y^2 - 5y$

(a) Gradient vector $\nabla f(x,y) =$
 $4x - 2y + 4$
 $6y - 2x - 5$

(b) Hessian matrix $H =$
 $[4, -2]$
 $[-2, 6]$

Eigenvalues of Hessian =
2.7639
7.2361

Conclusion: f is strictly convex because the Hessian is positive definite.

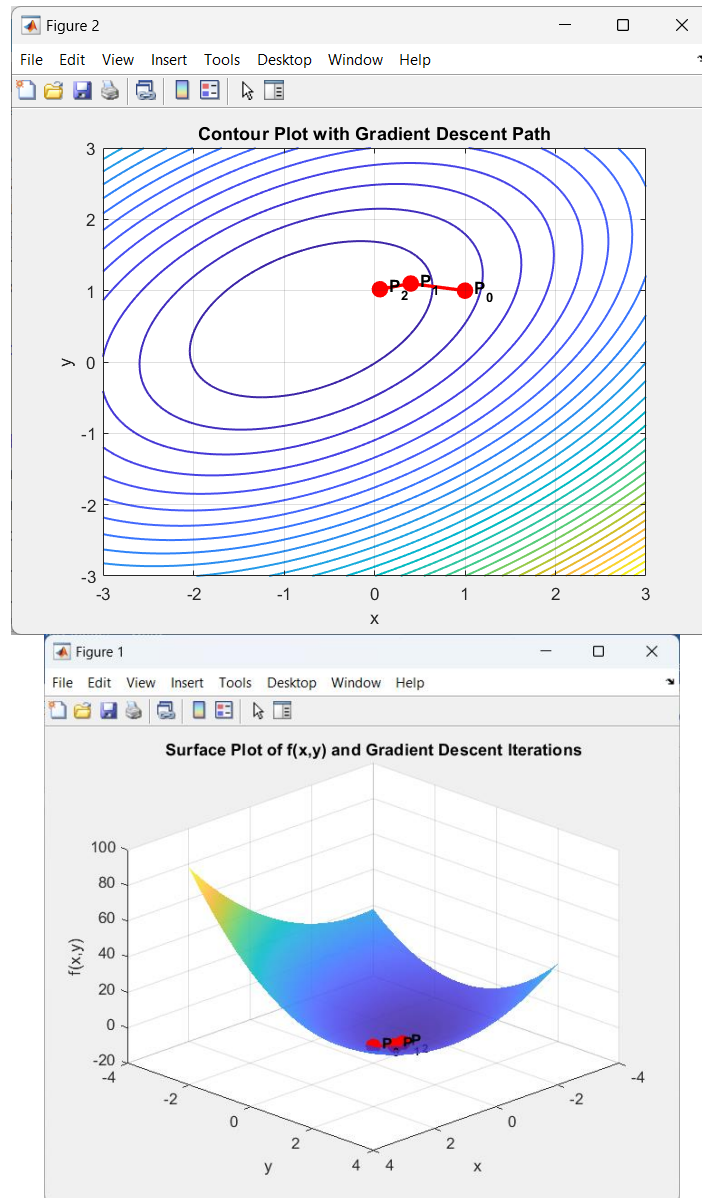
(c) Gradient Descent Iterations
Initial point: $(x_0, y_0) = (1.0000, 1.0000)$
Initial function value: $f(x_0, y_0) = 2.0000$

Iteration 1:
Current point = $(1.0000, 1.0000)$
Gradient at current point = $[6.0000; -1.0000]$
Update formula: $p_{\text{new}} = p - \alpha * \text{grad}_f(p)$
Numerical update: $p_{\text{new}} = [1.0000; 1.0000] - 0.10 * [6.0000; -1.0000]$
New point = $(0.4000, 1.1000)$
Function value at new point = -0.8300

Iteration 2:
Current point = $(0.4000, 1.1000)$
Gradient at current point = $[3.4000; 0.8000]$
Update formula: $p_{\text{new}} = p - \alpha * \text{grad}_f(p)$
Numerical update: $p_{\text{new}} = [0.4000; 1.1000] - 0.10 * [3.4000; 0.8000]$
New point = $(0.0600, 1.0200)$
Function value at new point = -1.8540

Summary of points and function values:

Iteration	x	y	f_xy
0	1	1	2
1	0.4	1.1	-0.83
2	0.06	1.02	-1.854



Problem 3:

Let $f(x, y) = x^2y$, and let $x(u, v) = u + v$, $y(u, v) = uv$. Define the composite function $g(u, v) = f(x(u, v), y(u, v))$. What is the value of $\frac{\partial g}{\partial u}$ at the point $(u, v) = (1, 1)$?

- A) 4
- B) 6
- C) 8
- D) 10

Solution

Correct Answer: C) 8

Explanation:

We use the chain rule for functions of two variables:

$$\frac{\partial g}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u}.$$

First, compute the partial derivatives of f :

$$\frac{\partial f}{\partial x} = 2xy, \frac{\partial f}{\partial y} = x^2.$$

Next, compute the partial derivatives of x and y with respect to u :

$$\frac{\partial x}{\partial u} = 1, \frac{\partial y}{\partial u} = v.$$

Now substitute into the chain rule:

$$\frac{\partial g}{\partial u} = (2xy) \cdot 1 + (x^2) \cdot v = 2xy + x^2v.$$

At $(u, v) = (1, 1)$, we have:

$$x = u + v = 2, y = uv = 1.$$

Then:

$$2xy = 2 \cdot 2 \cdot 1 = 4, x^2v = 4 \cdot 1 = 4.$$

Adding them gives $4 + 4 = 8$.

Thus, $\frac{\partial g}{\partial u}(1, 1) = 8$.

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Problem 4:

Consider a simple neuron in a neural network. It receives an input $x = 2$ and has a target output $y = 1$. The neuron computes a weighted sum $z = w \cdot x + b$, where w is the weight and b is the bias. The activation function is the sigmoid: $a = \sigma(z) = \frac{1}{1+e^{-z}}$. The loss function is the squared error: $L = (a - y)^2$. Initially, the parameters are $w = 0$ and $b = 0$. Using the chain rule, what is the value of the partial derivative $\frac{\partial L}{\partial w}$ at this initial point?

- A) -1.0
- B) -0.5
- C) 0.5
- D) 1.0

Solution

Correct Answer: B) -0.5

Explanation:

We need to compute $\frac{\partial L}{\partial w}$ using the chain rule. The dependencies are: L depends on a , a depends on z , and z depends on w . Thus,

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}.$$

Step 1: Compute $\frac{\partial L}{\partial a}$.

$L = (a - y)^2$, so

$$\frac{\partial L}{\partial a} = 2(a - y).$$

At the initial point, $w = 0, b = 0$, so $z = 0$, and $a = \sigma(0) = 0.5$. Given $y = 1$,

$$\frac{\partial L}{\partial a} = 2(0.5 - 1) = 2(-0.5) = -1.$$

Step 2: Compute $\frac{\partial a}{\partial z}$.

For the sigmoid function, $\frac{d\sigma}{dz} = \sigma(z)(1 - \sigma(z)) = a(1 - a)$. At $a = 0.5$,

$$\frac{\partial a}{\partial z} = 0.5 \times (1 - 0.5) = 0.5 \times 0.5 = 0.25.$$

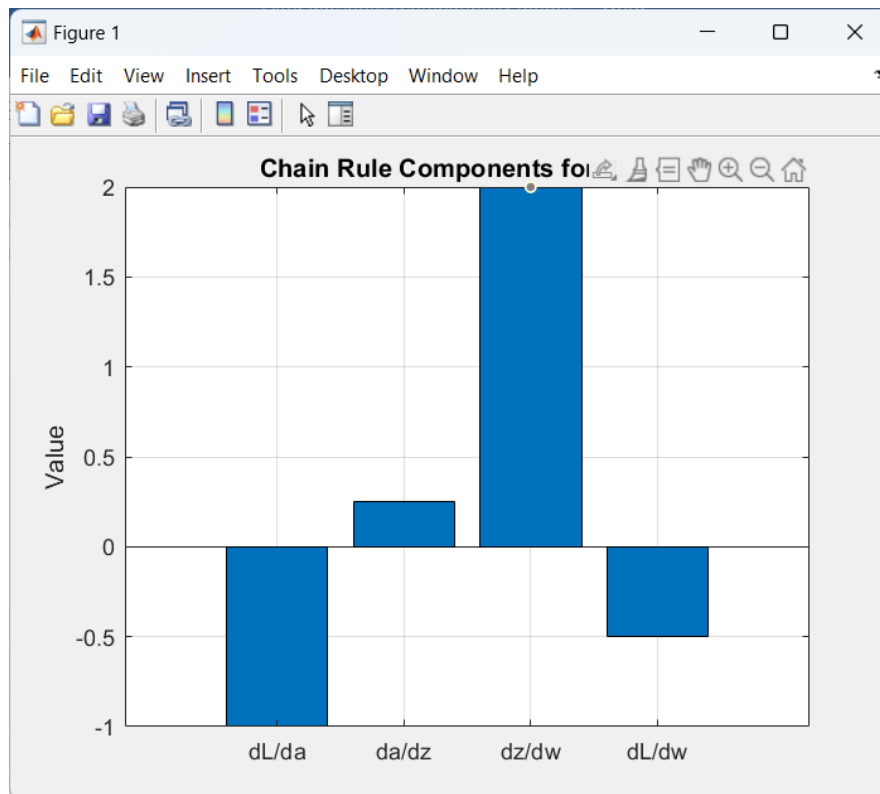
Step 3: Compute $\frac{\partial z}{\partial w}$.

$z = w \cdot x + b$, so $\frac{\partial z}{\partial w} = x = 2$.

Step 4: Multiply.

$$\frac{\partial L}{\partial w} = (-1) \times 0.25 \times 2 = -0.5.$$

Thus, the gradient is -0.5 . This value indicates that increasing w slightly would decrease the loss, as expected since the prediction $a = 0.5$ is below the target $y = 1$.



Problem 5:

Consider a small neural network with two layers. The input is $x = 2$. The first layer has weight $w_1 = 1$ and bias $b_1 = 0$, producing $z_1 = w_1x + b_1$. The activation function is ReLU: $a_1 = \max(0, z_1)$. The second layer has weight $w_2 = 0.5$ and bias $b_2 = 0$, producing $z_2 = w_2a_1 + b_2$. The output activation is sigmoid: $a_2 = \sigma(z_2) = \frac{1}{1+e^{-z_2}}$. The target output is $y = 1$, and the loss function is binary cross-entropy: $L = -[y \ln(a_2) + (1 - y) \ln(1 - a_2)]$.

What is the value of the partial derivative $\frac{\partial L}{\partial w_1}$ at this initial point?

- A) -0.5
- B) -0.26
- C) 0
- D) 0.5

Solution

Correct Answer: B) -0.26

Explanation:

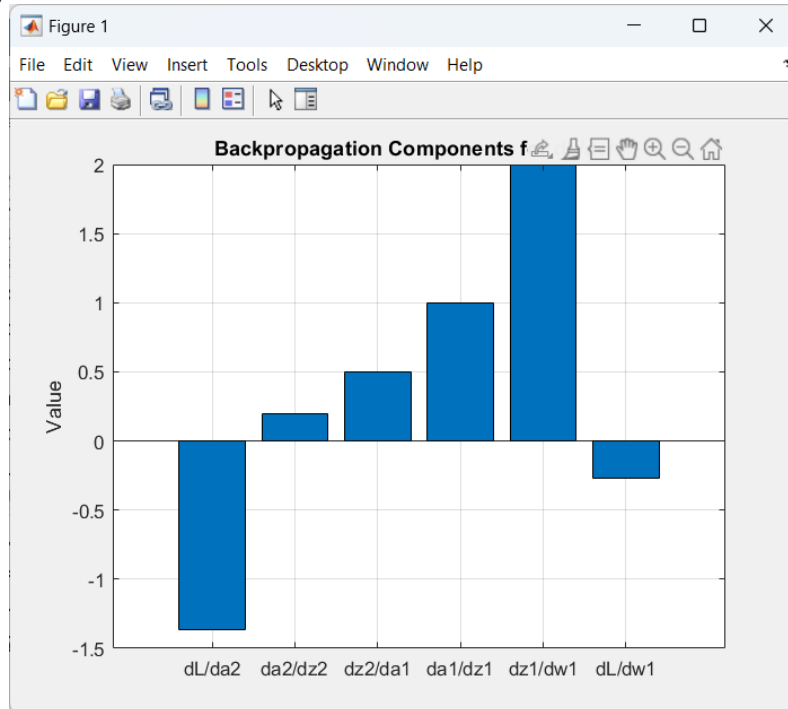
We need to compute $\frac{\partial L}{\partial w_1}$ using the chain rule through multiple layers. The dependencies are:

$$L \rightarrow a_2 \rightarrow z_2 \rightarrow a_1 \rightarrow z_1 \rightarrow w_1.$$

Thus,

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}.$$

Then, complete by yourself



Problem 6:

Consider a linear regression model with a single parameter w (no bias term). The original loss function (mean squared error) for a small dataset simplifies to:

$$J(w) = (w - 5)^2.$$

To prevent overfitting, a data scientist decides to apply **Elastic Net regularization** which combines L1 and L2 penalties. The regularized loss function is:

$$J_{\text{reg}}(w) = (w - 5)^2 + \lambda_1 |w| + \lambda_2 w^2,$$

with $\lambda_1 = 2$ and $\lambda_2 = 1$.

Assuming the optimal w is non-negative ($w \geq 0$), what is the value of w that minimizes the regularized loss function?

- A) $w = 1$
- B) $w = 2$
- C) $w = 3$
- D) $w = 4$

Solution

Correct Answer: B) $w = 2$

Explanation:

Given $w \geq 0$, we have $|w| = w$. The regularized loss becomes:

$$J_{\text{reg}}(w) = (w - 5)^2 + 2w + w^2.$$

Step 1: Expand the expression.

$$J_{\text{reg}}(w) = 2w^2 - 8w + 25$$

Step 2: Differentiate with respect to w .

$$\frac{dJ_{\text{reg}}}{dw} = 4w - 8$$

Step 3: Set derivative equal to zero and solve.

$$4w - 8 = 0 \Rightarrow 4w = 8 \Rightarrow w = 2$$

Step 4: Verify that $w = 2$ satisfies $w \geq 0$. Yes, it does.

Step 5: Check second derivative to confirm it's a minimum.

$$\frac{d^2J_{\text{reg}}}{dw^2} = 4 > 0$$

So $w = 2$ is indeed a minimum.

Thus, the optimal parameter value is $w = 2$.